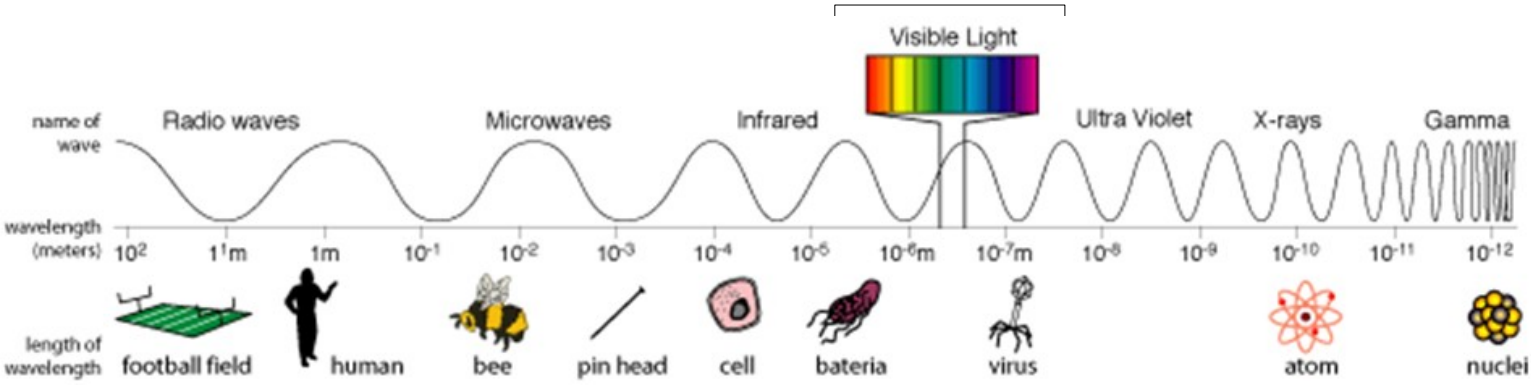
The Electromagnetic Spectrum

Go to the NASA website
http://science.hq.nasa.gov/kids/imagers/ems/index.html

1. Fill in the missing labels on the electromagnetic spectrum.



Click on *'What are electromagnetic waves?'*

2. In your own words, describe what electromagnetic waves are.

3. What did James Clerk Maxwell and Heinrich Hertz study?

Click on *'Different kinds of electromagnetic waves have different wavelengths.'*

4. In your own words, describe what is meant by wavelength. Include a diagram as part of your description.

Click on *'Radio waves'*

5. Radio waves have the *longest/shortest* wavelengths in the electromagnetic spectrum.

6. What length can radio waves be?

7. Give three uses for radio waves.

8. Radio telescopes are used to study planets, comets, stars and galaxies.
    a. What are astronomers able to learn by studying the radio waves from these objects?

    b. What is an advantage of radio astronomy?

Click on *'Next Shorter Wavelength'* (Microwaves)

9. Give the range of lengths of microwaves.

10. What are longer microwaves used for?

11. Why are microwaves good for transmitting information from one place to another?

12. What are shorter microwaves used for?

Click on *'Next Shorter Wavelength'* (Infrared)

13. Give the approximate sizes of infrared waves?

14. What are far infrared waves and how do we experience them?

15. What are near infrared waves and what do we use them for?

Click on *'Next Shorter Wavelength'* (Visible)

16. Each colour of visible light has a different wavelength.
    a. Which colour has the shortest wavelength?
    b. Which colour of light has the longest wavelength?
    c. What do all the colours of light make when they are seen together?

17. Name two sources of visible light.

Click on *'Next Shorter Wavelength'* (Ultraviolet)

18. What type of organisms can see ultraviolet light?

19. How do scientists classify UV light?

20. What does the Earth's atmosphere do to UV light and why have scientists developed a UV index?

Click on *'Next Shorter Wavelength'* (X-rays)

21. What happens to the energy of light as its wavelength decreases?

22. Why do we normally talk about x-rays in terms of energy rather than wavelength?

23. X-rays were first discovered by Wilhelm Conrad Roentgen in 1895. Describe how he discovered them and how he showed his findings to the public.

24. Describe how x-ray images are taken of parts of your body.

Click on *'Next Shorter Wavelength'* (Gamma rays)

25. Describe the wavelength and energy of gamma rays.

26. How are gamma rays dangerous to living organisms?

27. How do humans use gamma rays?

28. Where and how are gamma rays produced?